1) Silver oxide reacts with copper as shown: $Ag_2O + Cu \rightarrow 2Ag + CuO$

By using the M_r of each substances show that the equation obeys the law of conservation of mass.

M_r of Ag_2O = 232 M_r of Cu = 63.5

M_r of Ag = 108 M_r of CuO = 79.5

sum of M_r of all reatants = 232 + 63.5 = 295.5

sum of M_r of all products = 2(108) + 79.5 = 295.5

total of M_r of all reactants = total of M_r of all products

2) A solution is made by dissolving 5 g of potassium nitrate in 200 cm^3 of water.
Calculate the concentration of this solution in g/dm^3

concentration = $\frac{5}{\frac{200}{1000}}$ = 25 g/dm^3

3) How many atoms are there in 3.0 mol of copper? (Avogadro number = 6.02×10^{23} per mole)

number of atoms = $3.0 \times 6.02 \times 10^{23} = 1.806 \times 10^{24}$

4) What is the mass of 1.505×10^{23} molecules of H_2O? (Avogadro number = 6.02×10^{23} per mole)

mol H_2O = $\frac{1.505 \times 10^{23}}{6.02 \times 10^{23}}$ = 0.25 mol

mass H_2O = 18 x 0.25 = 4.5 g

5) Chromium can be made by reacting chromium oxide with aluminium.
What mass of chromium can be made from 30.4 g of chromium oxide?

$$Cr_2O_3 + 3Al \rightarrow 2Cr + Al_2O_3$$

moles Cr_2O_3 = $\frac{30.4}{152}$ = 0.20 mol

moles Cr = 2 x 0.20 = 0.40 mol

mass Cr = 52 x 0.40 = 20.8 g

6) 1.35 g of aluminium was reacted with 2.28 g of fluorine to form aluminium fluoride.

Determine the limiting reagent and then calculate the mass of aluminium fluoride that is formed.

$$2Al + 3F_2 \rightarrow 2AlF_3$$

moles Al = $\frac{1.35}{27}$ = 0.05 mol

moles F_2 = $\frac{2.28}{38}$ = 0.06 mol

0.05 moles of Al reacts with 0.075 moles of F_2 but there is only 0.06 mol of F_2 and so F_2 is the limiting reagent

0.06 moles of F_2 forms 0.04 mol of AlF_3

mass AlF_3 = 84 x 0.04 = 3.36 g

7) Lithium reacts with nitrogen to form lithium nitride.

A student did an experiment to determine the molar ratio in which lithium (Li) reacts with nitrogen (N_2).

The student found that 0.840 g of lithium (Li) reacts with 0.560 g of nitrogen (N_2)

a) Calculate the number of moles of lithium (Li) reacting.

moles Li = $\frac{0.840}{7}$ = 0.12 mol

b) Calculate the number of moles of nitrogen (N_2) reacting.

moles N_2 = $\frac{0.560}{28}$ = 0.02 mol

c) Calculate the molar ratio in simplest whole numbers of Li : N_2 in this reaction.

ratio of moles Li : mole N_2 = 0.12 : 0.02 = 6 : 1

d) Write the formula of the lithium nitride formed. **Li_3N**

e) Write a balanced equation for this reaction. **$6Li + N_2 \rightarrow 2 Li_3N$**

Area	Strength	To develop	Area	Strength	To develop	Area	Strength	To develop
Done with care and thoroughness			Which numbers are part of formula			Can show conservation of mass		
Shows suitable working			Can work out M_r			Can do reacting mass calculations		
Does not round too much			Work out moles from mass			Understands limiting reagents		
Can use sig figs			Can work out mass from moles			Can determine molar reaction ratios		
Gives units			Calculate concentration in g/dm^3			Can use ratios to determine a formula		
Can convert units			Use Avogadro constant					